Lecture 08 (B07): Brock–Mirman — Uncertainty and Numerical Integration

Residuals, expectations, Gauss–Hermite and monomial quadrature, training diagnostics

Simon Scheidegger

HEC, University of Lausanne

Brock–Mirman: Residual, Expectation, Loss

Pathwise residual used in the notebook (full-depreciation $\delta = 1$ benchmark):

$$G_t = 1 - \beta \frac{C_t}{C_{t+1}} \cdot \alpha z_{t+1} K_{t+1}^{\alpha-1},$$

with

$$K_{t+1} = s_t z_t K_t^{\alpha}, \qquad C_{t+1} = (1 - s_{t+1}) z_{t+1} K_{t+1}^{\alpha}.$$

Partial depreciation. The uncertainty notebook uses $\delta = 0.1$; the marginal productivity of capital then becomes $\alpha z_{t+1} K_{t+1}^{\alpha-1} + (1-\delta)$ and $K_{t+2} = z_{t+1} K_{t+1}^{\alpha} + (1-\delta) K_{t+1} - C_{t+1}$. The closed-form $\phi = \alpha\beta$ only holds at $\delta = 1$.

Training loss:

$$\ell_\rho = \frac{1}{N} \sum_{i=1}^{N} |G_t^{(i)}|^2.$$

- Simulate states (K_t, z_t) forward
- Draw one next-period shock z_{t+1} per state

Expectation Handling

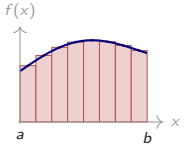
Brock–Mirman notebook: path averaging, one shock per state.

Richer models: approximate the conditional expectation explicitly via quadrature, see next three slides.

Numerical Integration: Why and How

Why an integral always shows up. Euler equations, Bellman operators, REE diagnostics, climate damages, structural moments, all reduce to evaluating $\mathbb{E}[g(\varepsilon)] = \int g(\varepsilon)\,\mu(\varepsilon)\,d\varepsilon$. Closed forms are rare, so we approximate by $\sum_q w_q\, g(\varepsilon_q)$. Two paradigms underlie every rule.

Deterministic: tile the area, integral $\to$ sum

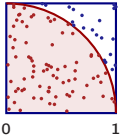


With $\Delta x = (b-a)/N$, $x_i = a + i\Delta x$, $I = \int_a^b f(x)\,dx \approx \sum_i w_i\, f(x_i)$:

- Midpoint: $I_N^{\mathbf{mid}} = \Delta x \sum_{i=1}^{N} f(x_i^{\mathbf{mid}})$, $\mathcal{O}(N^{-2})$
- Trapezoid: $I_N^{\mathbf{trap}} = \frac{\Delta x}{2}\left[f(x_0) + 2\sum_{i=1}^{N-1} f(x_i) + f(x_N)\right]$, $\mathcal{O}(N^{-2})$
- Simpson: parabolic arcs through triples, $\mathcal{O}(N^{-4})$
- Gauss–Hermite: optimal nodes/weights for e^{-x^2}, $\mathcal{O}(N^{-(2Q-1)})$

d-dim Cartesian grid: cost $\mathcal{O}(N^d)$, curse of dimensionality.

Random: throw darts to estimate π



Throw N uniform darts at $\Omega = [0,1]^2$:

$$\frac{\pi}{4} = \int_0^1\!\!\int_0^1 \mathbb{1}[x^2 + y^2 \le 1]\,dx\,dy \approx \frac{N_{\text{in}}}{N}$$

$$\widehat{\pi}_N = 4N_{\text{in}}/N, \quad \text{error } \mathcal{O}(N^{-1/2})$$

Independent of dimension, but slow.

Quadrature, I: Gauss–Hermite

Setup. Generic shock dimension d, with $\varepsilon' \sim \mathcal{N}(0, I_d)$. Brock–Mirman has $d = 1$ (one productivity shock); the IRBC chapter will use $d = N + 1$ (N country shocks plus one aggregate). We develop the framework here so it is ready for both cases.

One dimension. Gauss–Hermite uses Q nodes $\{\varepsilon_q\}$ (Hermite-polynomial roots) and weights $\{w_q\}$:

$$\mathbb{E}[h(\varepsilon)] \;=\; \int_{-\infty}^{\infty} h(\varepsilon)\,\frac{e^{-\varepsilon^2/2}}{\sqrt{2\pi}}\,d\varepsilon \;\approx\; \sum_{q=1}^{Q} w_q\, h(\varepsilon_q),$$

exact for univariate polynomials of degree $\leq 2Q - 1$.

Tensor product in d dimensions. Combine 1D rules along each axis:

$$\mathbb{E}\big[h(\boldsymbol{\varepsilon}')\big] \;\approx\; \sum_{q_1=1}^{Q}\cdots\sum_{q_d=1}^{Q}\Big(\textstyle\prod_j w_{q_j}\Big)\, h(\varepsilon_{q_1}, \ldots, \varepsilon_{q_d}).$$

- Strength: polynomial-exact in each coordinate, very accurate at low d. For $d = 1$ (Brock–Mirman) a $Q = 3$ rule is essentially exact.
- Weakness: cost Q^d blows up exponentially. For $Q = 3$, $d = 11$ (IRBC, $N = 10$): $\sim 1.8 \times 10^5$ nodes per state.

Quadrature, II: Monomial (Stroud-3) Rule

Idea. Drop full polynomial-exactness in every coordinate, keep exactness for total degree ≤ 3. Place nodes only on the principal axes of the d-dimensional shock space:

$$\boxed{\mathbb{E}\big[h(\varepsilon')\big] \;\approx\; \frac{1}{2d}\sum_{k=1}^{d}\Big[\,h\big(+\sqrt{d}\,\boldsymbol{e}_k\big) + h\big(-\sqrt{d}\,\boldsymbol{e}_k\big)\Big]}$$

where $\boldsymbol{e}_k$ is the k-th standard basis vector, giving $2d$ nodes at common radius $\sqrt{d}$. For Brock–Mirman ($d = 1$) this reduces to two nodes at ± 1; for IRBC ($d = N + 1$) it gives $2(N + 1)$ nodes.

What it integrates exactly

- $\mathbb{E}[\varepsilon_i'] = \mathbb{E}[\varepsilon_i'^{\,3}] = 0$ ($\pm$-symmetry).
- $\mathbb{E}[\varepsilon_i'\varepsilon_j'] = 0$, $i \neq j$ (each node lies on one axis).
- $\mathbb{E}[\varepsilon_i'^{\,2}] = 1$ (variance preserved).

Where it stops

- $\mathbb{E}[\varepsilon_i'^{\,4}]$ gives d vs. truth 3, a controlled fourth-order bias.

Why DEQNs love it

The integrand $h(\varepsilon')$ is the policy network composed with model primitives, smooth and only mildly nonlinear. Degree-3 exactness is enough for Euler-error tolerances of $\sim 10^{-3}$ on IRBC-class problems.

(Pichler 2011; Maliar & Maliar 2014)

Quadrature, III: Cost and Rules of Thumb

Per-state quadrature cost ($Q = 3$ for the tensor-product rule):

N	shocks $N{+}1$	tensor product 3^{N+1}	monomial $2(N{+}1)$	speed-up
2	3	27	6	$\sim 5\times$
5	6	729	12	$\sim 60\times$
10	11	177,147	22	$\sim 8{,}000\times$
20	21	$\sim 10^{10}$	42	$\sim 2\times 10^{8}\times$

Default rules of thumb

- $N{+}1 \leq 4$: tensor-product GH, $Q{=}5$.
- $5 \leq N{+}1 \lesssim 30$: Stroud-3 monomial.
- $N{+}1 \gtrsim 30$ or fat-tailed integrands: QMC (Sobol + inverse-CDF).

Practical notes

- Only the quadrature kernel changes; same DEQN loss, same training loop.
- Validate by tightening the rule and checking the Euler-error histogram is stable.
- Full derivations: lecture script §3.5.

Diagnostics: How Do We Know Training Worked?

Never trust a loss curve alone. On an independent *test* simulation of the trained policy, run these four complementary checks:

1. Euler-error distribution. Histogram $\log_{10}|e_{\mathrm{REE}}|$ on train and test states; both should peak near 10^{-3} or below, with no fatter right tail on test.
2. Policy-function benchmark. Overlay the DEQN policy on the analytic $s^\star = \beta\alpha$ for Brock–Mirman; for richer models, on a trusted reference (time iteration, perturbation).
3. Ergodic-set coverage. Scatter test states in the $(K_t, \log z_t)$ plane; the cloud should trace the training ergodic set — not collapse to a point, not spread like a grid.
4. Sanity checks. Verify accounting identities (budget, market clearing, non-negativity); check that ergodic moments (mean K, $\mathrm{std}(z)$, autocorrelations) match analytic or pre-computed values.

Green flags: low Euler error, matches benchmark, ergodic coverage, identities hold. Red flags: NaNs, fat tails, clamped policy, violated identities.

When no closed-form benchmark exists, draw inspiration from the VVUQ literature — Verification, Validation & Uncertainty Quantification (Roache 1998; Oberkampf & Roy 2010; ASME V&V 10/20) — which formalises residual convergence, solution verification, and error-bound reporting for computational models.